# Build a Real AI KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

**Word bank:** bigger fires negative positive

**1. How long by hand?** 240 cards, 300 passes, 1 minute per card.

240 × 300 = 72,000 cards = 72,000 minutes ≈ 1,200 hours ≈ 50 days

**2. Six steps to build an AI.** Number the steps in order, 1 to 6.

| | | | |
|---|---|---|---|
| 5 | Train: many passes, fix the weights | 1 | Get data |
| 3 | Split the data: training and test | 6 | Test, then use it |
| 2 | Ask a question | 4 | Build the network |

Why does the network never train on the test data? So we can check it on data it never saw.

**3. Read the network.**

A blue line is a positive weight. An orange line is a negative weight.

A thicker line is a bigger weight. A hidden neuron's square is blue where it fires.

**4. Predict, train, and test.** Write your prediction **before** each run.

| Run | Question | Hidden | My prediction (test %) | Training % | Test % |
|---|---|---|---|---|---|
| 1 | Fighter or Wizard? (2 numbers) | 4 | Answers vary. | about 90% | about 85% |
| 2a | Can it join the club? (2 numbers) | 0 | | about 60% | about 55% |
| 2b | Can it join the club? (2 numbers) | 2 | | about 96% | about 97% |
| 3 | Fighter or Wizard? (6 numbers) | 16 | | 100% | about 80% |
| 4 | Where does it live? (6 numbers) | 16 | | about 75% | about 35% |

Scores change a little with each random split. The patterns stay the same.

**Word bank:** data Giants line Minis overfitting

### 5. What did you see?

In Run 2a, the network had no hidden layer. It could only draw one line.

In Run 2b, one hidden neuron fires only for Giants. The other fires for everything except Minis.

In Run 3, the training score went up, but the test score went down. The network learned the training examples by heart. This is called overfitting.

In Run 4, the network wasn't better than guessing. Why? The home was picked at random. It isn't in the data.

An AI can only learn what is in its data.

### 6. Talk about it. What if someone used an AI like this to decide who gets a job?

A problem could be Answers vary. It only knows what is in its data; it can be wrong.

### Go Further (Day 2). Ask the network about 3 creatures from the cards.

| Creature | The network said | Really | Right? | Test creature? |
|---|---|---|---|---|
| Answers vary. | | | | |
| | | | | |
| | | | | |

*Fighter or Wizard?* (all 6 numbers). Test score with 0 hidden: 89% 4: 82% 16: 79%

Is the biggest network always the best? no Why? A big network can memorize the noise.